Regularization

- Lasso(L1)
- Ridge(L2)
- Elastic net

Lasso Regression

Think of **Lasso Regression** as a smart version of linear regression that **automatically removes unimportant features**.

Simple example
Imagine you're predicting a house's price using these features:

House size	Number of bedrooms	Age of the house	Distance to school	Wall color	Owner's favorite movie
….	….	….	….	….	….
….	….	….	….	….	….
….	….	….	….	….	….
….	….	….	….	….	….

Clearly, **wall color** and **owner's favorite movie** probably don't help predict the price.
A normal linear regression might consider these feature for model building.
However Lasso will ignore such features.

Lasso Regression says:
"If a feature isn't useful, I'll make its slope exactly **0** and ignore it."

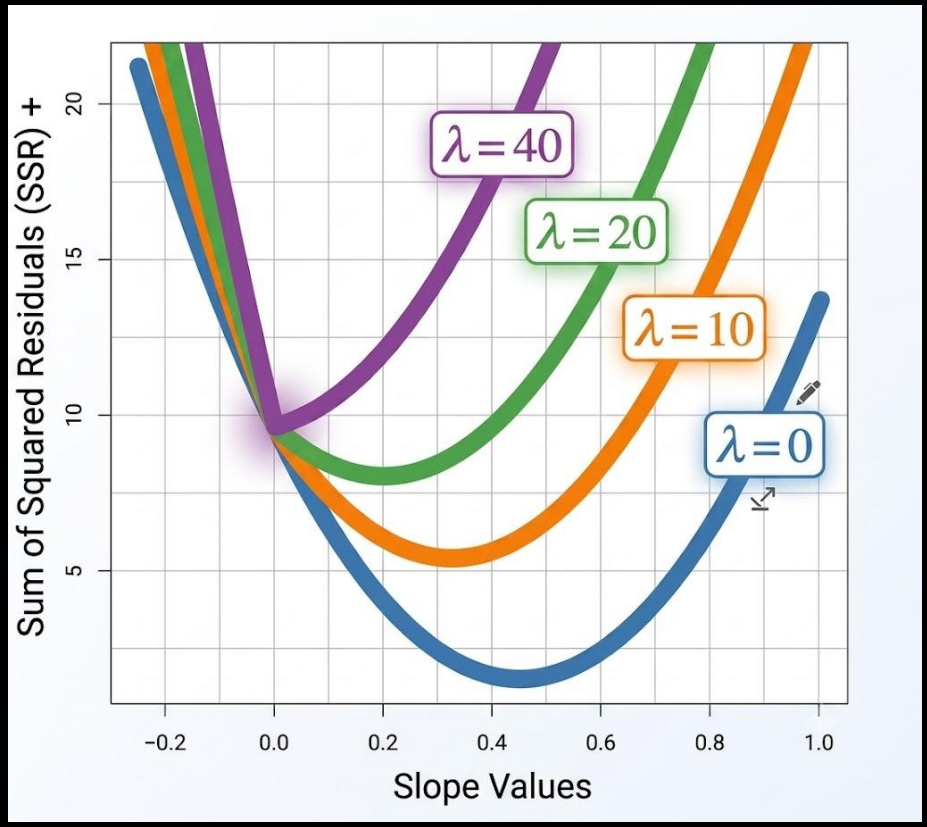
Lasso Regression: (L1 Regularization):
Lasso is used to do the **feature selection.**
Lasso adds a penalty term.

$$\lambda \sum_{i=1}^{m} |slope|$$

This penalty term is the sum of the absolute values of the coefficients(slope) which is multiplied by a constant often denoted as **lambda** or alpha. The equation of the same is given below.

Equation after adding the Lasso(L1)

$$(m, c) = \frac{1}{2m} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2 + \lambda \sum_{i=1}^{m} |slope|$$



The main aim of the Lasso is to reduce the **number of features** it is used to do the **feature selection.**

Ridge Regression

Think of **Ridge Regression** as a one more smart version of linear regression that **keeps all features in training but reduces their importance**.

Ridge Regression says:
"I'll keep all the features, but I'll make the slope smaller towards 0 does not make exactly to 0
so the model doesn't rely too heavily on any one feature."

Ridge Regression:(L2 Regularization)

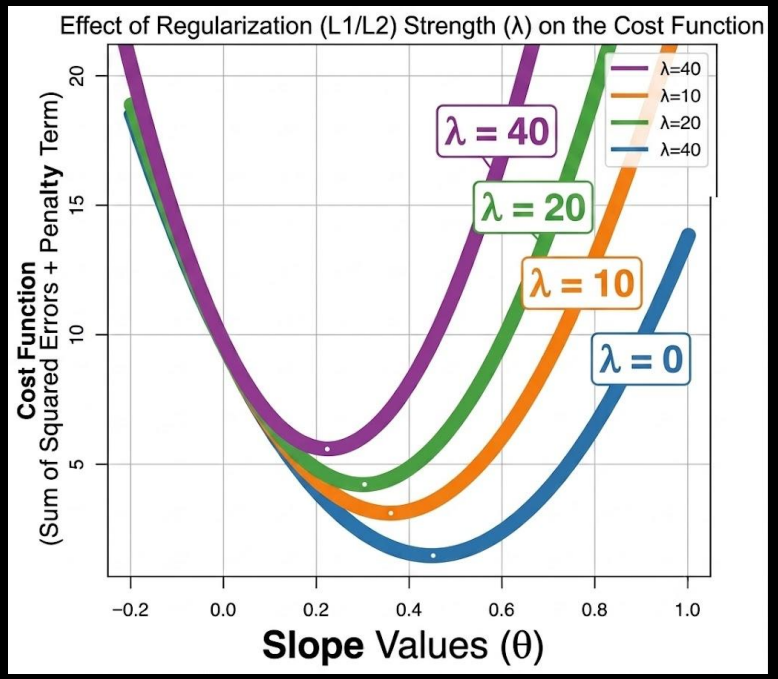
The main purpose of the Ridge is to reduce **overfitting**

When you want to reduce the impact of all features without eliminating them entirely.

$$+ \lambda \sum_{i=1}^{m} (slope)^2$$

Equation after adding the Ridge(L2)

$$(m, c) = \frac{1}{2m} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2 + \lambda \sum_{i=1}^{m} (slope)^2$$



Elastic Net Regression:

Elastic Net regression is an extension of linear regression that combines both L1 (lasso) and L2 (ridge) regularization techniques. It aims to overcome some limitations of ridge regression and lasso regression by providing a balance between feature Selection and overfitting.

Elastic Net strikes a balance between the two techniques and is useful for high-dimensional datasets with correlated features, offering flexibility in handling multicollinearity while performing feature selection.

$$j(\theta_o, \theta_1) = \frac{1}{2m}\sum_{i=1}^{m}(y_i - \hat{y_i})^2 + \lambda\sum_{i=1}^{m}(slope)^2 + \lambda\sum_{i=1}^{m}|slope|$$

An easy analogy
Imagine five people are carrying a heavy table.

Linear Regression: Some people might carry almost all the weight.

Ridge Regression: Everyone still helps, but the load is shared more evenly.

Lasso Regression: If someone isn't helping much, they are asked to step away completely (their weight becomes 0).

Overfitting:

If training Data has Very Good Accuracy while Testing Data has Bad Accuracy then this condition is called overfitting.

To solve such a problem need to perform **hyper-parameter tuning, OR Regularization .**

Under-fitting :

If training Data has Very Bad Accuracy, while Testing Data has Bad or Good Accuracy then this condition is called Under fitting.

To solve such problem we need to do the retraining of the model.

Overfilling and Under-fitting (Bias and Variance):

Bias :

Accuracy of the training data is mentioned by Bias.

High Bias : If Training Data has Bad Accuracy then.

Low Bias : if Training Data has Very **Good Accuracy**

Variance:

Accuracy of the test data is mentioned by Variance.

High Variance: If Testing Data Bad or Good Accuracy.

Low Variance: If Testing Data Very Good Accuracy.

Overfitting:

If training Data has Very Good Accuracy(Low Bias) while Testing Data has Bad Accuracy (High Variance) then this condition is called overfitting. To solve such a problem need to perform **hyper-parameter tuning.**

Under-fitting :

If training Data has Very Bad Accuracy(High Bias), while Testing Data has Bad (High Variance) or Good Accuracy (Low Variance) then this condition is called Under fitting.